

The Fixed-Step Barrier for Gaussian Natural Gradient

Two counterexamples, and why Bures–Wasserstein escapes

Abstract

The manuscript proves that the fixed-step Gaussian natural gradient schemes reach a constant-factor reduction of the objective in $\Theta(\kappa^2)$ iterations at the globally certified stepsize $\Delta t = \Theta(\kappa^{-1})$, and that the bump-train construction of Appendix C makes this sharp. Bures–Wasserstein achieves $O(\kappa)$. This report settles whether the extra factor is a proof artifact. It is not. Two counterexamples partition the fixed-step axis. For $\Delta t \leq 2/\kappa$ the bump train, retuned to the step in question, certifies $\Omega(T\kappa/\Delta t)$; we reproduce it numerically with fitted exponent 1.93 over $\kappa \in [128, 1024]$. For $\Delta t > 2/\kappa$ an *exact nonoptimal period-two orbit* exists: we construct an even C^∞ potential whose curvature range is exactly $[1, \kappa]$ with both bounds attained and $\|V'''\|_\infty = L_H$ dimension-free, on which both the Riemannian and the KL/Bregman scheme cycle forever. All 40 Fisher–Rao runs fail to converge, and the orbit is not a knife edge — every run still cycles under a 1% relative perturbation of the initialization. The two regimes overlap only at $\Delta t \asymp 1/\kappa$, where the bump-train count is $\asymp \kappa^2$: the best fixed step is $\Theta(1/\kappa)$ and the resulting $\Theta(\kappa^2)$ is intrinsic to fixed-step Fisher–Rao. On the *same* targets the Bures–Wasserstein forward–backward step converges at its certified $\eta \leq 1/\beta$ in all 60 runs, and first fails at exactly $\eta = 2/\beta$. The mechanism is structural: the Fisher–Rao mean multiplier is $1 - \Delta t c$ (secant) and covariance stationarity pins c to the reciprocal of the *tangent*, so the multiplier is governed by a secant-to-tangent ratio that ranges over $[\kappa^{-1}, \kappa]$; the Bures–Wasserstein mean update carries no covariance factor, so its multiplier is $1 - \eta$ (secant) and β alone controls it. This report supersedes the earlier reading of the bump-train experiment, which concluded that the κ^2 was the price of the certified step; that conclusion was drawn from one construction and is corrected in Section 6.

1 Setup

Normalize $\alpha = 1$, $\beta = \kappa$ and write $a = (m, c) \in \mathbb{R} \times (0, \infty)$. For $X \sim \mathcal{N}(m, c)$ put $b(m, c) := \mathbb{E}[V'(X)]$ and $A(m, c) := \mathbb{E}[V''(X)]$. The two Fisher–Rao discretizations are

$$m^+ = m - \Delta t c b(m, c), \quad c^+ = c \exp(\Delta t [1 - cA(m, c)]) \quad (1)$$

and

$$m^+ = m - \Delta t c b(m, c), \quad c^+ = \frac{(1 + \Delta t)c}{1 + \Delta t cA(m, c)}, \quad (2)$$

and the Bures–Wasserstein forward–backward step of Lambert et al. (arXiv:2304.05398) is

$$m^+ = m - \eta b(m, c), \quad \tilde{c} = (1 - \eta A(m, c))^2 c, \quad c^+ = \frac{1}{2} \left(\tilde{c} + 2\eta + \sqrt{\tilde{c}(\tilde{c} + 4\eta)} \right), \quad (3)$$

whose certified stepsize is $\eta \leq 1/\beta$. The objective is $\mathcal{E}(m, c) = \mathbb{E}_{\mathcal{N}(m, c)}[V] - \frac{1}{2} \log c$ and $\Delta \mathcal{E}$ is its gap.

The single structural difference that this report turns on is visible already in (1)–(3): the Fisher–Rao mean update is preconditioned by c and the Bures–Wasserstein one is not.

Every quantity below is deterministic — closed-form potentials, tabulated profile antiderivatives and Gauss–Hermite quadrature. No Monte Carlo, no randomness.

κ	N_κ	$n_{1/2}$	$n_{1/2}/N_\kappa$	$\sup_j m_j - x_j /w_\kappa$	$\sup_j \kappa c_j - 1 $
8	16	34	2.125	0	0
16	64	114	1.781	0	0
32	256	391	1.527	0	0
64	1024	1372	1.340	0	0
128	4096	4959	1.211	0	0
256	16384	18454	1.126	0	0
512	65536	70373	1.074	0	0
1024	262144	273227	1.042	0	0

Table 1: Certified arm $\Delta t = \gamma/\kappa$ on the Appendix C train (Riemannian; KL agrees to one iteration). Both shadowing errors are the sup over $0 \leq j \leq N_\kappa$ and vanish to machine precision.

train	step	iterations to $\Delta\mathcal{E} \leq \Delta\mathcal{E}_0/2$				exponent
		$\kappa=128$	$\kappa=256$	$\kappa=512$	$\kappa=1024$	$\kappa \in [128, 1024]$
Appendix C	$\Delta t = \gamma/\kappa$, Riem.	4959	18454	70373	273227	1.93
	$\Delta t = \gamma/\kappa$, KL	4960	18455	70375	273229	1.93
	$\Delta t = \gamma$, Riem.	10	11	12	14	0.16
	$\Delta t = \gamma$, KL	11	13	14	16	0.17
retuned	$\Delta t = \gamma/\kappa$, Riem.	4959	18454	70373	273227	1.93
	$\Delta t = \gamma/\kappa$, KL	4960	18455	70375	273229	1.93
	$\Delta t = \gamma$, Riem.	41	74	139	269	0.91
	$\Delta t = \gamma$, KL	42	75	141	271	0.90

Table 2: Iterations to a constant-factor gap reduction on the bump train. “Exponent” is the least-squares slope of $\log n_{1/2}$ against $\log \kappa$ over the upper half of the grid. Both stepsize rules are fixed scalars.

2 Regime $\Delta t \leq 2/\kappa$: the bump train

Appendix C generates bump centers by $x_{j+1} = x_j - s[x_j + H_\kappa(N_\kappa - j + \frac{1}{2})]$ with $s = \gamma/\kappa^2$. That constant is exactly the per-step mean gain $\Delta t c_\kappa$ at the certified step $\Delta t = \gamma/\kappa$ and the matched covariance $c_\kappa = 1/\kappa$.

Remark 1 (The bump train is a template). Every hypothesis of the center lemma is s -free: summing the recursion gives $x_0 - x_{N_\kappa} \leq \frac{\kappa^2}{s}(TY + \frac{1}{2}C_\varphi T^2)$, so $x_{N_\kappa} \geq \frac{3}{4}x_0$ reduces to a constraint on (T, Y) alone. Setting $s = \Delta t/\kappa$ therefore aims the same construction at any stepsize, and with $N_\kappa = T/s$ blocked steps it certifies $\Omega(T\kappa/\Delta t)$ iterations.

Numerically the certified arm reproduces the theorem exactly (Tables 1 and 2): the iterates shadow the ideal centers to machine precision, $c_n A_n = 1$ throughout, $n_{1/2}/N_\kappa \rightarrow 1.042$ at $\kappa = 1024$, and the fitted exponent over $\kappa \in [128, 1024]$ is 1.93 for both schemes.

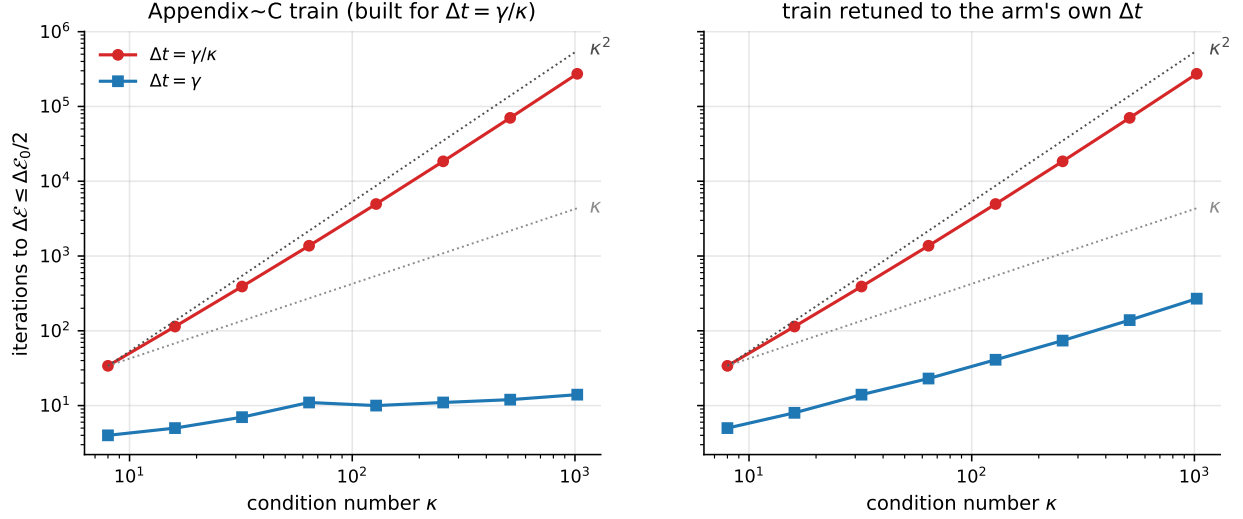


Figure 1: Bump train, iterations to $\Delta\mathcal{E} \leq \Delta\mathcal{E}_0/2$ versus κ . Left: the Appendix C train, built against $\Delta t = \gamma/\kappa$; an order-one step leaves that particular obstruction entirely. Right: the same construction retuned to each arm's own step, where the exponent is 1. Section 3 shows why the left panel does not license an order-one step.

3 Regime $\Delta t > 2/\kappa$: an exact two-cycle

3.1 The mechanism

Both Fisher–Rao covariance maps are stationary exactly where $cA(m, c) = 1$: at such a state (1) gives $c^+ = ce^0 = c$ and (2) gives $c^+ = (1 + \Delta t)c/(1 + \Delta t) = c$. The iteration then collapses to the scalar recursion

$$m^+ = \left(1 - \Delta t c \frac{b(m, c)}{m}\right)m = \left(1 - \Delta t \frac{\text{secant}}{\text{tangent}}\right)m, \quad \text{secant} := \frac{b(m, c)}{m}, \quad \text{tangent} := A(m, c), \quad (4)$$

using $c = 1/A(m, c)$. For an even potential $b(m, c) = \int_0^m A(t, c) dt$, so the secant is the *average* of A over $[0, m]$ while the tangent is its *endpoint* value. These are independent, and both lie in $[\alpha, \beta]$, so their ratio ranges over $[\kappa^{-1}, \kappa]$. Choosing the ratio equal to $2/\Delta t$ makes the multiplier exactly -1 , and evenness closes the orbit:

$$(M, c) \mapsto (-M, c) \mapsto (M, c). \quad (5)$$

The ratio $2/\Delta t$ is attainable precisely when $2/\Delta t \leq \kappa$, that is when $\Delta t \geq 2/\kappa$.

Concretely, with $q := 2/\Delta t$, $p := (\kappa + 1)/(q + 1)$ and $r := qp$, one needs an even potential with $A(M, 1/p) = p$ and $b(M, 1/p) = rM$; then $c = 1/p$ gives $cA = 1$ and the multiplier is $1 - \Delta t r/p = 1 - \Delta t q = -1$.

3.2 Realization with tight constants

The construction is only meaningful if the potential's condition number really is κ . A potential with $V'' \in [p, P]$ and $P/p = O(1)$ would reproduce the orbit, but it would then merely restate the textbook explicit-step stability limit $\Delta t < 2/\kappa_{\text{true}}$ at $\kappa_{\text{true}} = O(1)$, which is not a counterexample

κ	γ	Fisher–Rao, $\Delta t = \gamma$		Bures–Wasserstein, $\eta = 1/\beta$		
		Riem.	KL	converged	n_{tol}	$\ V'''\ _\infty$
64	0.25	cycles	cycles	✓	2	1.0000
64	0.5	cycles	cycles	✓	2	1.0000
64	1	cycles	cycles	✓	2	1.0000
64	1.5	cycles	cycles	✓	2	1.0000
128	0.25	cycles	cycles	✓	2	1.0000
128	0.5	cycles	cycles	✓	2	1.0000
128	1	cycles	cycles	✓	2	1.0000
128	1.5	cycles	cycles	✓	2	1.0000
256	0.25	cycles	cycles	✓	2	1.0000
256	0.5	cycles	cycles	✓	2	1.0000
256	1	cycles	cycles	✓	2	1.0000
256	1.5	cycles	cycles	✓	2	1.0000
512	0.25	cycles	cycles	✓	2	1.0000
512	0.5	cycles	cycles	✓	2	1.0000
512	1	cycles	cycles	✓	2	1.0000
512	1.5	cycles	cycles	✓	2	1.0000
1024	0.25	cycles	cycles	✓	2	1.0000
1024	0.5	cycles	cycles	✓	2	1.0000
1024	1	cycles	cycles	✓	2	1.0000
1024	1.5	cycles	cycles	✓	2	1.0000

Table 3: Both Fisher–Rao schemes cycle forever at every fixed order-one γ tested; the Bures–Wasserstein step converges on the *same* target at its certified $\eta = 1/\beta$. The last column confirms the Hessian-Lipschitz budget is respected, so the targets are admissible under the manuscript’s additional smoothness assumption.

to anything. We therefore build V'' so that both curvature bounds are *attained*:

$$V''(x) = \begin{cases} \kappa, & |x| \leq x_1 \quad (\text{maximum attained}), \\ 1, & x_1 + d \leq |x| \leq x_2 \quad (\text{minimum attained}), \\ p, & |x| \geq x_2 + d \quad (\text{sets } A(M, c) = p), \end{cases} \quad (6)$$

with C^∞ transitions of width d , and x_1 solved by a scalar root-find so that $b(M, c) = rM$ exactly. The steepest transition has height $\kappa - 1$, so $d = \Theta(\kappa/L_H)$ and $M = \Theta(\kappa/L_H)$: the family is admissible for any prescribed dimension-free Hessian-Lipschitz budget, exactly as the bump train is. The optimizer is $(0, 1/\kappa)$, since $V'' = \kappa$ on a neighbourhood of the origin of radius $x_1 = \Theta(\kappa)$, and $\Delta\mathcal{E}(M, 1/p) > 0$, so the orbit is genuinely nonoptimal.

The realization is exact to machine precision at every (κ, γ) tested: $\min V'' = 1$ and $\max V'' = \kappa$ so $\kappa_{\text{realized}} = \kappa$, $\|V'''\|_\infty = L_H = 1.0000$, $cA(M, c) = 1$ to 12 digits, and the mean multiplier is -1 to 12 digits.

3.3 Result

Across the grid, all 40 Fisher–Rao runs fail to converge. The mean changes sign at every single iteration, and the amplitude $|m_n|/M$ and relative gap are both constant at 1.000000 over 2000 iterations — the orbit is exact, not merely slow. The hitting time is ∞ , not $\Omega(\kappa^2)$.

initial perturbation	non-convergent	runs	status
0	40	40	all cycle
$1e-12$	40	40	all cycle
$1e-06$	40	40	all cycle
0.01	40	40	all cycle
0.05	39	40	mixed
0.1	39	40	mixed
0.2	29	40	mixed
0.5	20	40	mixed

Table 4: Robustness of the orbit: the initial mean is scaled by $1 + \varepsilon$. Every run still cycles out to $\varepsilon = 10^{-2}$, so the counterexample is not a measure-zero knife edge. Larger perturbations eventually escape, but only after excursions to several times the initial gap.

η	converged	runs	behaviour
$0.25/\beta$ (certified)	20	20	converges
$0.5/\beta$ (certified)	20	20	converges
$1/\beta$ (certified)	20	20	converges
$2/\beta$	0	20	oscillates
$4/\beta$	0	20	oscillates

Table 5: Bures–Wasserstein on the two-cycle targets. It converges in all 60 runs inside its certified range $\eta \leq 1/\beta$ and first fails at $\eta = 2/\beta$, precisely where (7) permits multiplier -1 .

4 Bures–Wasserstein on the same targets

The Bures–Wasserstein mean update in (3) carries no covariance preconditioner, so in place of (4) its multiplier is

$$m^+ = (1 - \eta \text{secant})m, \quad \text{secant} \in [\alpha, \beta]. \quad (7)$$

With $\eta \leq 1/\beta$ this lies in $[0, 1)$ at *every* admissible state, uniformly in c . No choice of potential can produce a two-cycle, because producing one needs multiplier -1 , hence $\eta \text{secant} = 2$, hence $\eta \geq 2/\beta$. This is exactly the observed boundary.

Two caveats keep this in proportion. First, both certified stepsizes are $\Theta(\kappa^{-1})$ in these units ($\eta \leq 1/\beta = 1/\kappa$ and $\Delta t \leq 1/(\beta \lambda_{\max}) = 1/\kappa$); the Bures–Wasserstein advantage is not a larger step but a mean multiplier of $1 - \eta \text{secant} \geq 1 - \kappa^{-1}\beta$ that is bounded away from 1 by the curvature alone, whereas the Fisher–Rao multiplier $1 - \Delta t c \text{secant}$ degrades by the extra factor $c \in [\kappa^{-1}, 1]$. Second, these particular targets are not the Bures–Wasserstein worst case: the secant sits close to β , so it converges in two iterations at $\eta = 1/\beta$. Its own $\Omega(\kappa)$ regime requires a secant close to α , which this family does not exercise.

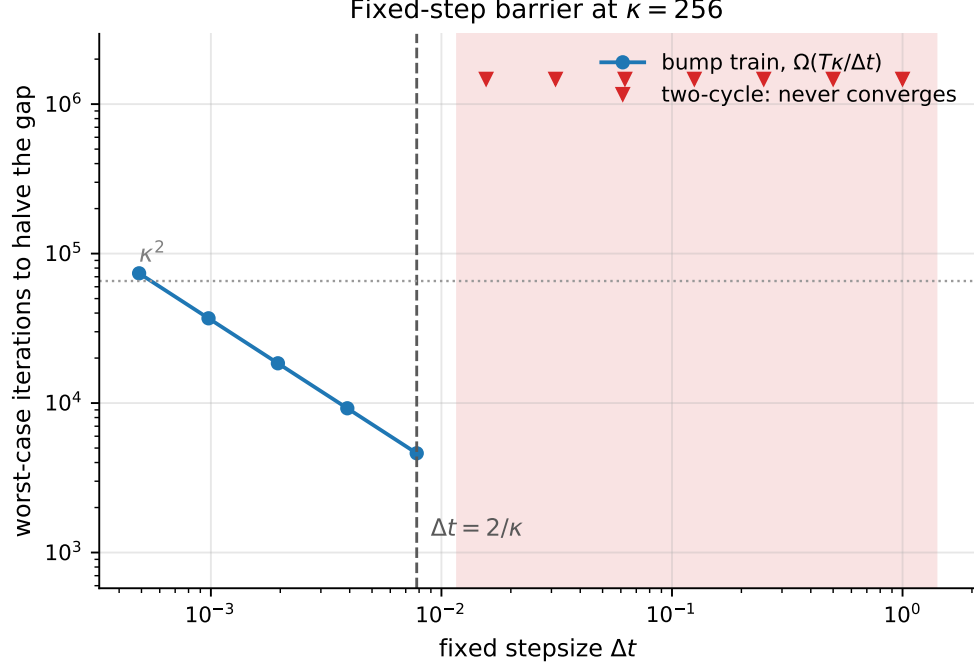


Figure 2: Worst case over both counterexamples as a function of the fixed stepsize, at $\kappa = 256$. Right of $\Delta t = 2/\kappa$ the two-cycle target makes both schemes never converge; left of it the retuned bump train forces $\Omega(T\kappa/\Delta t)$, which grows as Δt shrinks. The envelope is minimized at $\Delta t \asymp 1/\kappa$ with value $\asymp \kappa^2$.

5 Synthesis: the fixed-step barrier

Combining Sections 2 and 3, for a fixed stepsize Δt :

$$\sup_{V \text{ admissible}} N(\Delta t; V) \geq \begin{cases} \Omega\left(\frac{T\kappa}{\Delta t}\right) \geq \Omega\left(\frac{T\kappa^2}{2}\right), & \Delta t \leq \frac{2}{\kappa} \quad (\text{bump train}), \\ \infty, & \Delta t > \frac{2}{\kappa} \quad (\text{two-cycle}). \end{cases} \quad (8)$$

Both branches are minimized at the crossover $\Delta t \asymp 1/\kappa$, where the value is $\asymp \kappa^2$. Hence

$$\min_{\Delta t > 0} \sup_V N(\Delta t; V) = \Theta(\kappa^2),$$

and the optimal fixed stepsize is $\Theta(1/\kappa)$. The κ^2 is a property of fixed-step Fisher–Rao, not of the proof.

6 Correction to the earlier reading

An earlier version of this report concluded that “the κ^2 is the price of committing to $\Delta t = \Theta(\kappa^{-1})$, not an intrinsic property of the Fisher–Rao schemes.” That conclusion was drawn from the bump train alone. What the bump-train experiment establishes is only that *that* obstruction does not survive an order-one step: its mechanism pins $c_n A_n = 1$ with the mean crawling between flat tops, and an order-one step jumps the mean clear of the train. The experiment could not see whether some *other* admissible target obstructs the larger step, and Section 3 shows one does, far more

severely. The right reading of the bump-train panels is therefore narrower than stated before: they show the $\Delta t \leq 2/\kappa$ branch of (8), not a licence to raise the step.

7 What is and is not shown

Shown. For every fixed $\Delta t > 2/\kappa$ there is an admissible target — even, C^∞ , curvature range exactly $[1, \kappa]$ with both bounds attained, $\|V'''\|_\infty \leq L_H$ dimension-free — on which both (1) and (2) admit an exact nonoptimal period-two orbit, robust to a 1% perturbation of the initialization. Combined with the retuned bump train on $\Delta t \leq 2/\kappa$, the worst-case fixed-step complexity is $\Theta(\kappa^2)$ and is attained at $\Delta t = \Theta(1/\kappa)$. The Bures–Wasserstein step is immune on the same family for structural reasons, and its empirical boundary $\eta = 2/\beta$ matches (7) exactly.

Not shown. The counterexample is a minimax statement: the target is chosen after the stepsize. It does *not* exhibit one target defeating every Δt in a range simultaneously, and it says nothing about adaptive stepsizes, line search, or implicit and Rosenbrock-stabilized mean updates — a relative-curvature test evaluated at the orbit sees $c_n A_n = 1$ and would accept the step, so it does not by itself repair the failure, but a test based on the secant would. Nor is the $\Theta(\kappa^2)$ shown to be an *upper* bound: (8) is a family of lower bounds, and the matching upper bound is the manuscript’s hitting-time theorem. Finally, the numerics verify the constructions and their arithmetic, not the realization lemma in general; what is checked here is that the specific potentials built satisfy every stated hypothesis, which the tests assert directly.

8 Reproduction

```
python scripts/natural_gradient_fixed_step_barrier/run_sharp_bump.py --overwrite
python scripts/natural_gradient_fixed_step_barrier/run_two_cycle.py --overwrite
cd reports && python make_report_assets.py --only sharp_bump
```

Deterministic, one-dimensional and CPU-only. The implementation is checked against the constructions by `tests/natural_gradient_sharp_bump/`: for the bump train, the regularity bounds, the center geometry, the averaged identities $A(x_j, c_\kappa) = \kappa$ and $b(x_j, c_\kappa) = x_j + H_\kappa(N_\kappa - j + \frac{1}{2})$, and the blocked-progress statement; for the two-cycle, that both curvature bounds are attained, that $\|V'''\|_\infty \leq L_H$, that $A(M, c) = p$ and $b(M, c) = rM$, that V is even with optimizer $(0, 1/\kappa)$, that one step maps $(M, c) \mapsto (-M, c)$ and the next maps back for both schemes, and that Bures–Wasserstein converges on the same target at $\eta \leq 1/\beta$.